# Brazilian E-Commerce Sales Analysis

Week 1 Final Report: Data Cleaning, Sales Analysis and Visualization

Data Science Internship, WeIntern Pvt Ltd
Author: Janarthanan M
Date: 12 November 2026

## 1. Title and Objective

This report documents the Week 1 assignment. The objective is to clean the Olist Brazilian e-commerce dataset, document every cleaning decision, calculate business KPIs, analyse sales trends and customer behaviour, and communicate the findings through clear, labelled charts and a dashboard mockup. All numbers in this report are produced by the notebook *notebooks/week1_analysis.ipynb*, which runs from top to bottom, and the KPI formulas were double-checked against the raw files.

**Headline result:** delivered orders generated R$13,221,498 from 96,478 orders (average order value R$137.04). Revenue grew strongly through 2017 and early 2018, but only 3.0% of customers bought more than once, and late deliveries are linked to much lower review scores.

## 2. Dataset Overview

The project uses the public Olist Brazilian E-Commerce dataset (Kaggle). Eight related tables were used. The geolocation table was not needed for the required analysis and is not included. Raw files are kept unchanged in *data/raw/*.

| Table | Rows | Columns | Business meaning |
|---|---|---|---|
| customers | 99,441 | 5 | Customer key, city and state |
| orders | 99,441 | 8 | Order status and purchase / delivery timestamps |
| order_items | 112,650 | 7 | One row per item sold, with price and freight |
| payments | 103,886 | 5 | Payment method, installments and value |
| reviews | 99,224 | 7 | Review score and comments |
| products | 32,951 | 9 | Product category and size |
| sellers | 3,095 | 4 | Seller city and state |
| category_translation | 71 | 2 | Portuguese to English category names |

Every column was classified as an identifier, numeric, categorical, date or free-text field (saved in *outputs/reports/column_dictionary.csv*). All eight date columns were stored as text and were converted to real dates. The dataset has no quantity column: each *order_items* row is one unit, and Olist hides product names, so products are identified by ID and category.

## 3. Cleaning Summary

Cleaning followed the handbook workflow: inspect, measure, decide, explain. Missing values were never filled with invented dates. Rows were flagged rather than deleted, and the raw data was preserved.

## Missing values

| Table.column | Missing | % | Strategy |
|---|---|---|---|
| reviews.review_comment_title | 87,656 | 88.3 | Keep missing |
| reviews.review_comment_message | 58,247 | 58.7 | Keep missing |
| orders.order_delivered_customer_date | 2,965 | 3.0 | Keep missing + flag |
| products.product_name_lenght | 610 | 1.9 | Keep missing |
| products.product_category_name | 610 | 1.9 | Fill with label 'unknown' |
| products.product_description_lenght | 610 | 1.9 | Keep missing |
| products.product_photos_qty | 610 | 1.9 | Keep missing |
| orders.order_delivered_carrier_date | 1,783 | 1.8 | Keep missing + flag |
| orders.order_approved_at | 160 | 0.2 | Keep missing + flag |

*Two product dimension columns with only 2 missing rows each (0.006%) were filled with the median. The order date gaps follow order status (for example canceled or shipped orders have no delivery date), so they are real states, not errors.*

## Duplicates

No fully duplicated rows exist in any of the eight tables, so nothing was removed. Repeated IDs were investigated and are valid: orders with several items repeat the order ID in *order_items*, split payments repeat it in *payments*, and some orders have more than one review. The pair (review_id, order_id) is unique, so no accidental copies were found.

## Validation checks

28 validation rules were tested; 19 passed with zero violations. The rules covered price and freight signs, payment values, review score range, date formats and date order, standardized status and payment values, category spelling, state codes, and links between tables. The rules that found something are listed below, with the action taken.

| Rule | Table | Violations | Action |
|---|---|---|---|
| Payment value must be greater than 0 | payments | 9 | Kept and flagged: zero-value vouchers and 'not_defined' payments carry no money |
| Payment installments must be at least 1 | payments | 2 | Kept and flagged |
| Payment type has a defined value | payments | 3 | Kept and flagged (value is 0) |
| Categories missing an English translation | products | 2 | Translated manually: pc_gamer, portateis_cozinha_e_preparadores_de_alimentos |
| Carrier pickup earlier than approval | orders | 1,359 | Kept and flagged: timestamps conflict but the order itself is valid |
| Customer delivery earlier than carrier pickup | orders | 23 | Kept and flagged |
| Delivered order without a delivery date | orders | 8 | Flagged and excluded from delivery-time metrics |
| Delivery date present but order not delivered | orders | 6 | Kept; status is the source of truth |
| Order has no item rows | orders | 775 | Kept in order counts; excluded from revenue because there is no price |

## Cleaning decisions

| Issue | Evidence | Decision | Why |
|---|---|---|---|
| Date columns stored as text | 8 date columns across 3 tables | Converted to datetime | Needed for monthly trends and delivery time; 0 values failed to parse |
| Missing order timestamps | 160 / 1,783 / 2,965 gaps in approval / carrier / customer delivery | Kept as missing | Gaps follow order status (canceled, shipped, unavailable); filling would invent dates |
| Delivered orders without a delivery date | 8 orders | Flagged, excluded from delivery-time metrics only | They still count as completed sales |
| Conflicting delivery timestamps | 1382 orders where pickup precedes approval or delivery precedes pickup | Flagged, kept | Order value and status are valid; deleting would remove real sales |
| Products without category | 610 products, 1,603 order items | Labelled 'unknown' | Deleting them would remove revenue and bias category results |
| Two categories missing English names | pc_gamer, portateis_cozinha_e_preparadores_de_alimentos | Translated manually | Avoids Portuguese labels in charts |
| Misspelled column names | product_name_lenght, product_description_lenght | Renamed to *_length | Consistent column names |
| Zero-value payments | 9 rows (vouchers and 'not_defined') | Flagged, kept | Valid zero-cost vouchers; no effect on price-based revenue |
| Duplicate rows | 0 fully duplicated rows in all 8 tables | Nothing removed | Repeated IDs are valid multi-item orders or split payments |
| Orders without item rows | 775 orders (mostly unavailable / canceled) | Kept in the orders table | No price exists, so they add no revenue |
| Zip codes lost leading zeros | Stored as integers | Padded to 5 characters | Zip prefixes are identifiers, not numbers |
| Review comment text | Large and mostly empty | Dropped from the cleaned reviews file only | Not analysed; the raw file keeps the text |

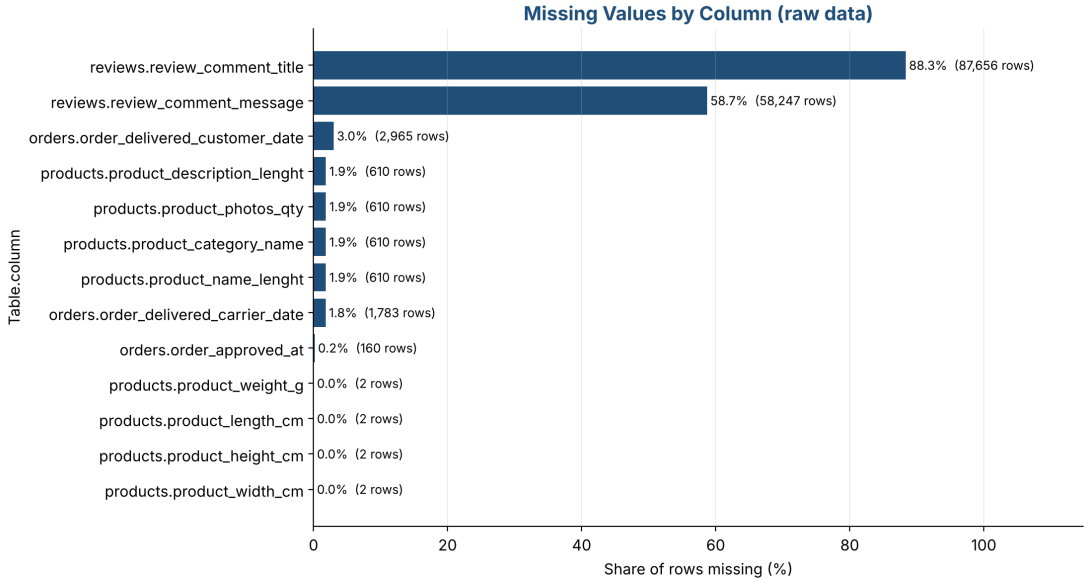



***Figure 1.*** *Share of missing values by column in the raw data. Review comments are the largest gap because customers often skip them; the order date gaps match order status.*

Cleaned files are saved separately in *data/cleaned/*, including *final_sales_dataset.csv* (one row per sold item, with English category names).

## 4. KPI Summary

KPIs use **completed sales**: items in orders with status *delivered*. Revenue is the sum of item price and excludes freight. A unit is one item row. The formulas were verified with an independent calculation on the raw files, and delivered payments reconcile to price plus freight within 0.02%.

| KPI | Value | Meaning |
|---|---|---|
| Total Revenue | R$13,221,498.11 | Total value of all completed (delivered) sales, excluding freight |
| Total Orders | 96,478 | Number of delivered orders |
| Average Order Value | R$137.04 | Revenue divided by total orders |
| Total Units Sold | 110,197 | Sum of items sold in delivered orders |
| Top-Selling Category | bed_bath_table (10,953 units) | Category with the most units sold |
| Top Revenue Category | health_beauty (R$1,233,132) | Category with the highest sales contribution |
| Top Revenue-Generating Product | bb50f2e236... (health_beauty, R$63,560) | Product with the highest revenue (Olist hides product names) |
| Revenue Realization | 97.3% | Delivered product value as a share of the value of all orders |

Only 97.3% of the product value of all orders sits in delivered orders; the remainder is canceled, unavailable or still in progress. The best-selling category by units is bed_bath_table (10,953 units), while health_beauty earns the most revenue (R$1,233,132), so volume and revenue leaders differ.

## 5. Trend Analysis

Trend statistics use the complete months January 2017 to August 2018, because 2016 and September to October 2018 contain only a handful of orders.

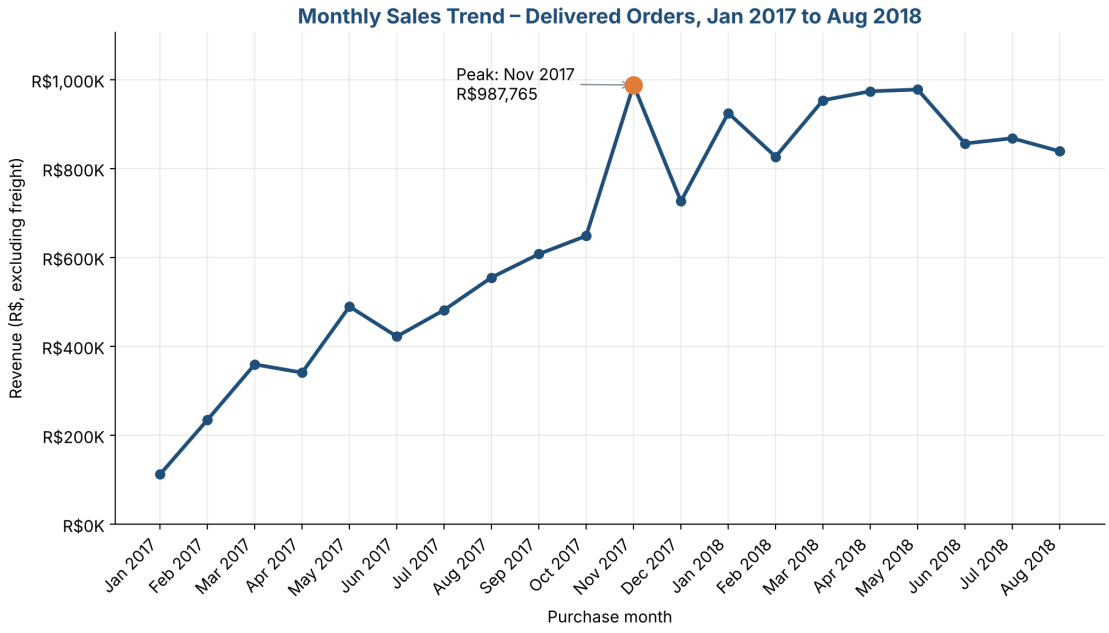



***Figure 2.*** *Revenue grew from R$111,798 in Jan 2017 to R$838,577 in Aug 2018. Nov 2017 is the peak (+52% on Oct 2017), and 1,176 orders on 24 Nov 2017 alone (6.1x a typical November day) point to a promotion such as Black Friday as the likely driver. Revenue then fell 26% in December, so the jump was a spike, not a new baseline.*

Revenue grew 650% between Jan 2017 and Aug 2018. The peak month is Nov 2017 (R$987,765, +52% on Oct 2017). Looking at single days, 1,176 orders were placed on 24 November 2017 against a typical

November day of about 192. This points to a promotion such as Black Friday, but the data has no campaign information, so it is an inference. Revenue then fell 26% in December, which shows the jump was a spike and not a new baseline. Revenue stayed between about R$830K and R$980K per month from January to August 2018, so growth flattened after the first half of the period.

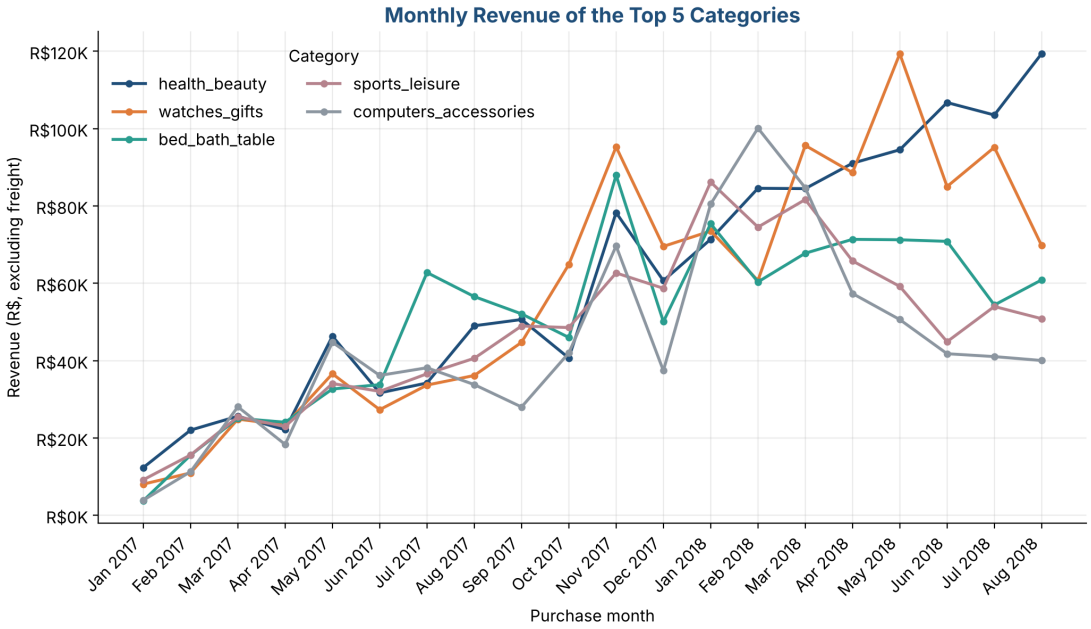



***Figure 3.*** *Between Jan-Aug 2017 and Jan-Aug 2018, watches_gifts grew fastest (+242%) while bed_bath_table grew slowest (+109%), so category growth is uneven.*

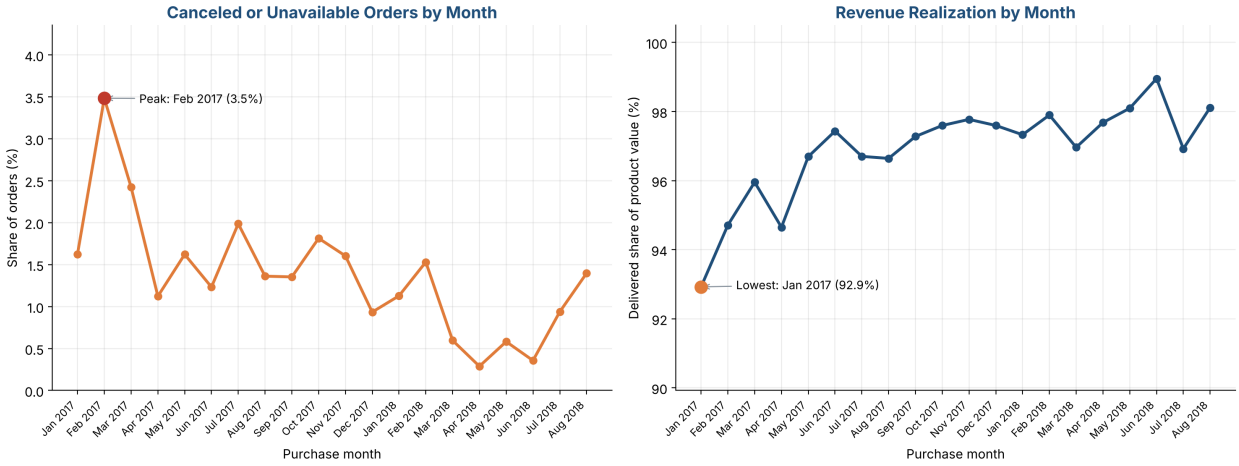



***Figure 4.*** *Canceled or unavailable orders fell from an average of 1.9% of monthly orders in the first six months to 0.7% in the last six (peak 3.5% in Feb 2017), and revenue realization rose from 95.4% to 97.8% (lowest 92.9% in Jan 2017). This suggests fulfilment became more reliable over time, so less ordered value was lost; the data cannot show the reason.*

Region differences are covered in Section 6 and in the charts of Section 7. Delivered orders hold 97.3% of all product value, and 1.24% of orders were canceled or unavailable, worth R$97,243 of product value.

## 6. Customer Insights

| Segment | Customers | Share of customers | Share of revenue | Revenue per customer |
|---|---|---|---|---|
| One-time customers | 90,557 | 97.00% | 94.49% | R$137.96 |
| Repeat customers (2+ delivered orders) | 2,801 | 3.00% | 5.51% | R$260.05 |

**Which customers order most often?** Repeat customers are only 3.00% of 93,358 customers, but each one is worth R$260 against R$138 for a one-time customer. The repeat rate is highest in RJ (3.28%) and lowest in CE (1.67%) among states with 1,000 or more customers.

**Is behaviour evenly distributed?** No. The top 10% of customers generate 41.1% of revenue.

**Where is order density highest?** SP contributes 38.3% of revenue and the top three states 63.4%. Sao Paulo holds 15.6% of orders and the top 10 cities 35.1%.

**Are high-value orders concentrated in certain cities?** High-value orders (the top 10%, from R$269) follow order volume: Sao Paulo has 13.5% of them against 15.6% of all orders, so no city is dominated by expensive orders.

**Do customer groups prefer different categories?** Repeat customers spend a larger share on bed_bath_table (+5.6 percentage points) and a smaller share on health_beauty (-2.0 points) than one-time customers.

**Delivery and satisfaction.** Average delivery takes 12.6 days (median 10.2) and 8.11% of orders arrive after the estimated date. Late orders average a review score of 2.57 against 4.29 for on-time orders. MA has the highest late rate (19.7%).

## 7. Visualization Section

All charts have a title, labelled axes, one consistent colour system (blue for normal values, orange to highlight the leader, red for the worst case) and a short written finding. The set contains six bar charts, three line charts (Figures 2, 3 and 4), one pie chart and a dashboard mockup.

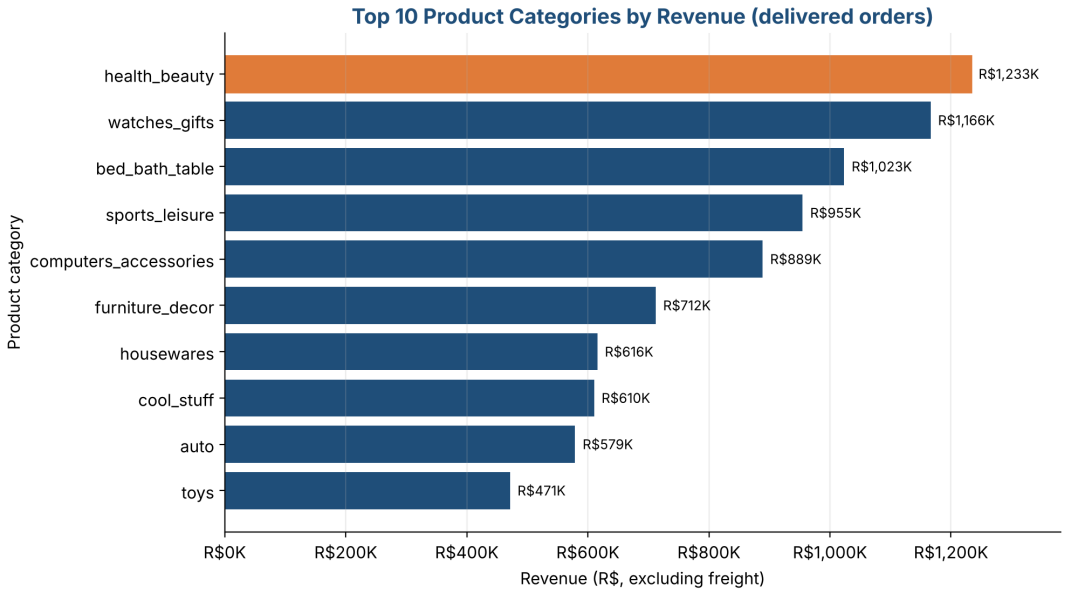



***Figure 5.*** *health_beauty earns the most (R$1,233,132). The top 10 categories together provide 62.4% of revenue, so a small set of categories drives results.*

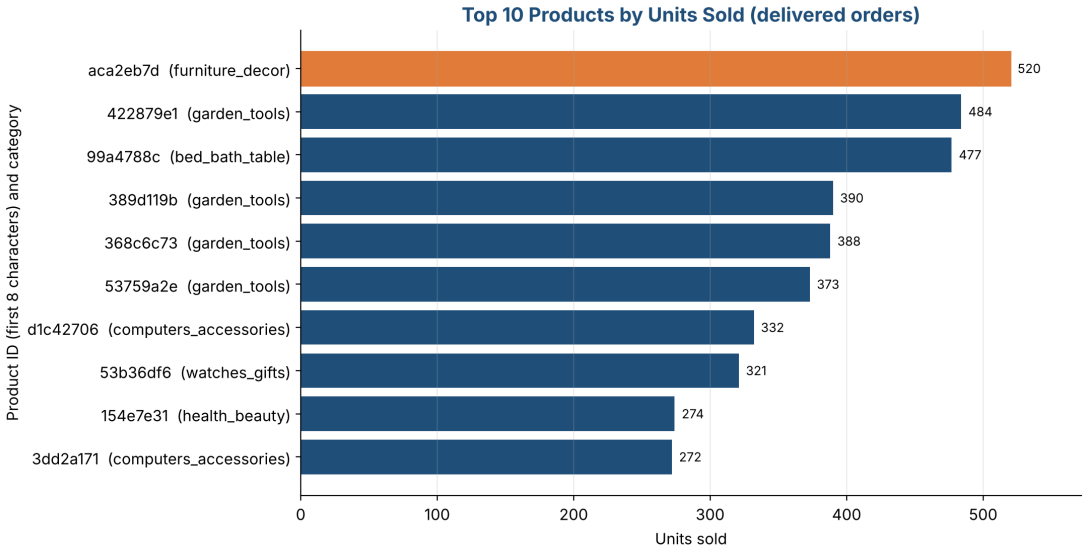



***Figure 6.*** *The best-selling product (furniture_decor) sold 520 units. 8 of these 10 products cost less than the average item price (R$120), and together they are only 3.5% of all units, so volume is spread across many products rather than a few hits.*

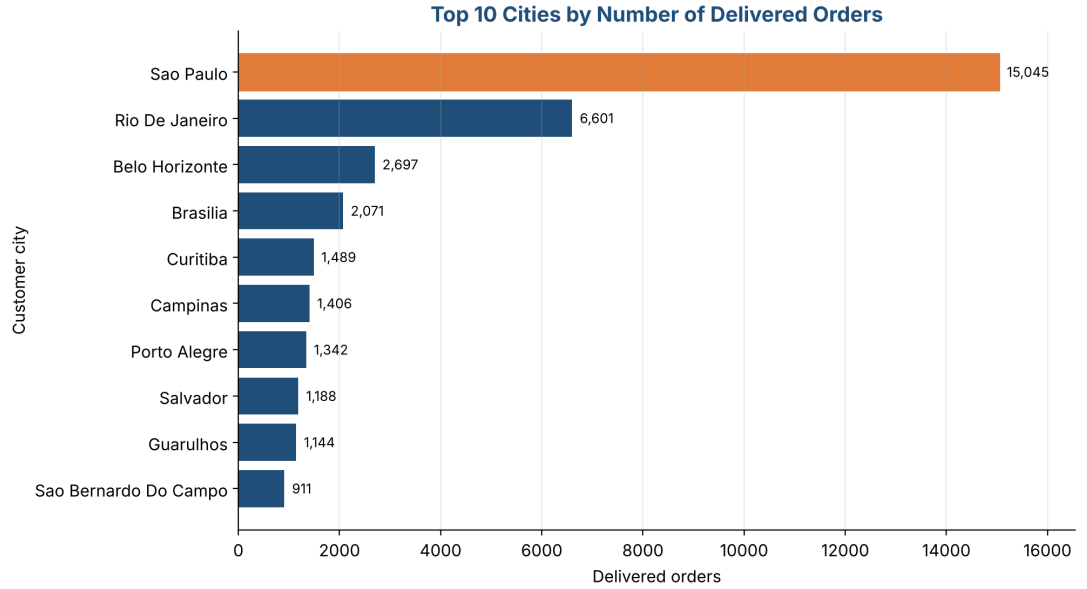



***Figure 7.*** *Sao Paulo alone accounts for 15.6% of delivered orders and the top 10 cities for 35.1%, so order density is highly concentrated in a few large cities.*

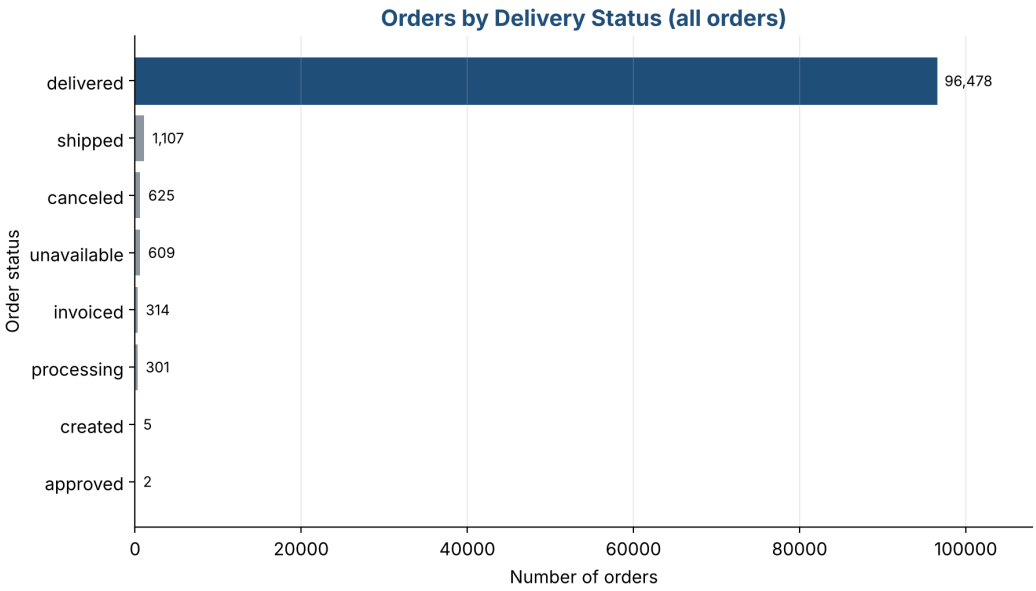



***Figure 8.*** *97.0% of orders reach the delivered status. Canceled and unavailable orders together are 1.24% of orders (R$97,243 of product value that never became revenue).*

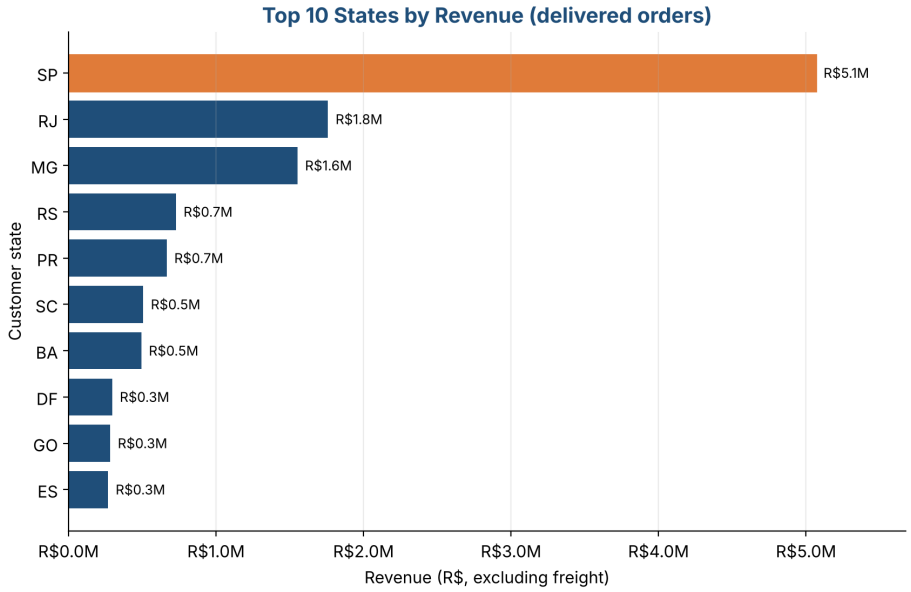



***Figure 9.*** *SP contributes 38.3% of revenue and the top three states 63.4%, so service quality in these states has the largest effect on overall results.*

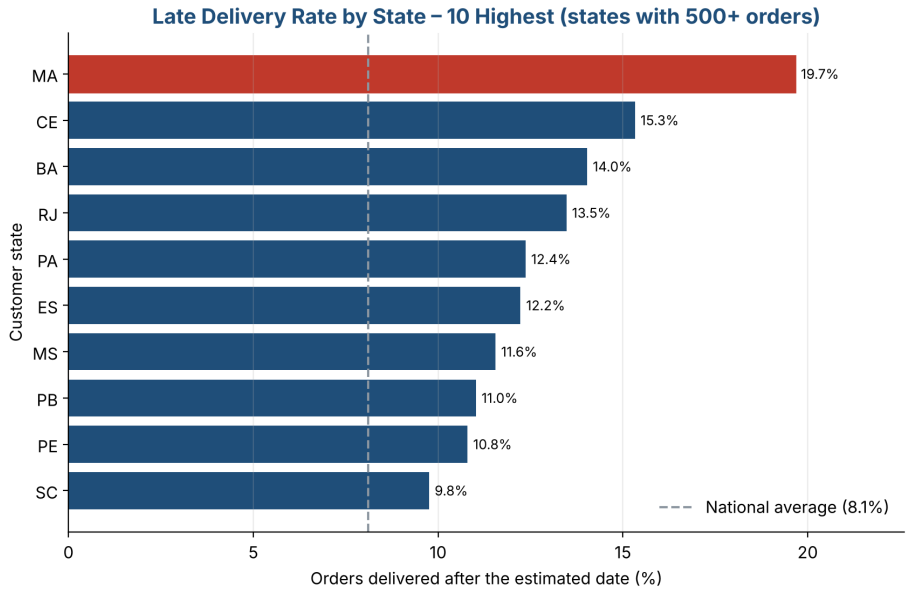



***Figure 10.*** *MA has the highest late rate (19.7% versus 8.1% nationally). Late orders average a review score of 2.57 compared with 4.29 for on-time orders, which suggests lateness is linked to unhappy customers.*

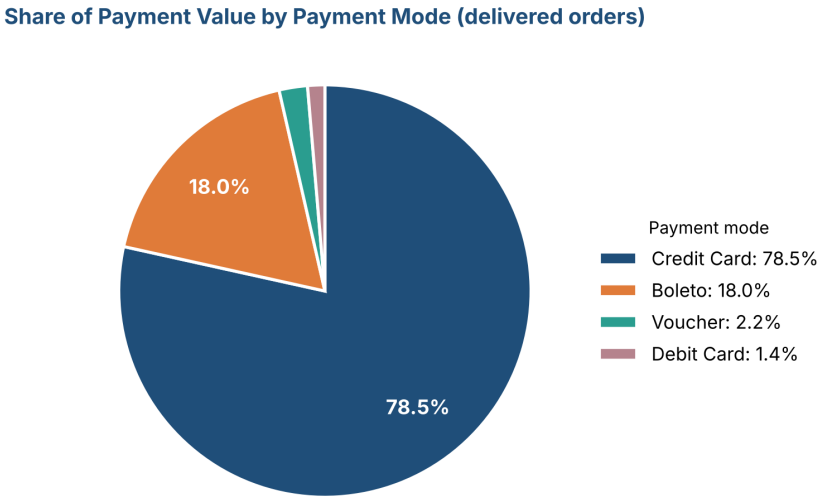



***Figure 11.*** *Credit Card accounts for 78.5% of payment value, and 67% of card payments are split into installments, so installment-friendly checkout matters to customers.*

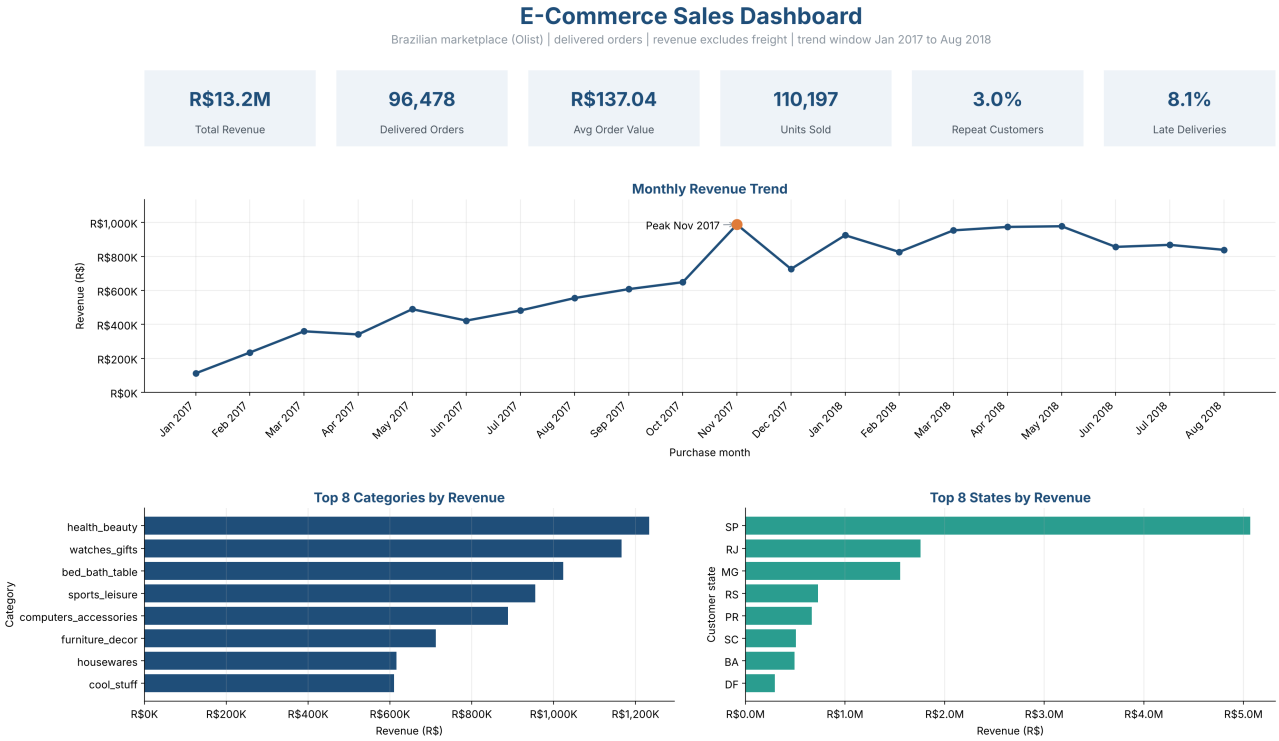



***Figure 12.*** *One page combines six KPIs, the monthly trend, category and regional comparisons: SP and health_beauty lead, and Nov 2017 stands out as the peak month.*

# 8. Key Findings

Each finding is an observation supported by data, followed by why it matters. Recommendations are kept separate in the next section.

## 1. Overall performance

**Observation:** Delivered orders generated R$13,221,498 from 96,478 orders, an average order value of R$137.04, and delivered orders hold 97.3% of the product value of all orders (the rest is canceled, unavailable or still in progress).

**Why it matters:** These figures are the baseline against which any future campaign should be judged.

## 2. Growth with a one-off spike

**Observation:** Monthly revenue rose 650% from Jan 2017 to Aug 2018. Nov 2017 was the peak month (+52% on Oct 2017), with 1,176 orders on 24 Nov 2017 alone (6.1x a typical November day), and December fell 26% below the peak.

**Why it matters:** A one-day surge that fades suggests a promotion such as Black Friday rather than lasting growth; this is an inference, not proof.

## 3. Category concentration

**Observation:** health_beauty leads revenue (R$1,233,132) while bed_bath_table leads units (10,953). The top 10 categories provide 62.4% of revenue.

**Why it matters:** Revenue and volume leaders differ, so marketing should follow revenue while stock planning follows units.

### 4. Weak customer retention

**Observation:** Only 3.00% of 93,358 customers ordered more than once, yet they produce 5.51% of revenue (R$260 per repeat customer versus R$138 for one-time customers).

**Why it matters:** Repeat customers are worth more individually, so even a small rise in repeat buying would lift revenue.

### 5. Skewed customer value

**Observation:** The top 10% of customers generate 41.1% of revenue.

**Why it matters:** Revenue depends on a minority of high-spending customers, so protecting them matters.

### 6. Geographic concentration

**Observation:** SP contributes 38.3% of revenue and the top three states 63.4%. Sao Paulo holds 15.6% of orders. High-value orders (R$269 and above) follow order volume rather than clustering in one city: Sao Paulo has 13.5% of them against 15.6% of all orders.

**Why it matters:** A few regions carry the business, so their delivery quality has an outsized effect, but there is no sign of a city that only buys expensive items.

### 7. Delivery quality

**Observation:** Average delivery takes 12.6 days and 8.11% of orders arrive after the estimated date; MA is the worst state at 19.7%. Late orders score 2.57 in reviews against 4.29 for on-time orders.

**Why it matters:** Late delivery is associated with clearly lower satisfaction, which can reduce repeat purchases (association, not proven cause).

### 8. Payments and cancellations

**Observation:** Credit Card makes up 78.5% of payment value and 67% of card payments use installments. 1.24% of orders were canceled or unavailable (R$97,243 of product value). Monthly cancellations fell from an average of 1.9% in the first six months to 0.7% in the last six, and revenue realization rose from 95.4% to 97.8%.

**Why it matters:** Checkout should be designed around card installments, and the falling cancellation rate suggests fulfilment became more reliable, though the data cannot show why.

## 9. Recommendations

**Launch retention campaigns (finding: weak retention)**

Target one-time buyers with follow-up offers after delivery, because each repeat customer is worth R$260 against R$138 for a one-time customer.

**Prepare capacity for promotional peaks (finding: growth with a one-off spike)**

Plan stock and logistics ahead of November campaigns and keep the post-peak decline in mind when forecasting.

**Fix delivery in the weakest states first (finding: delivery quality)**

Review carrier performance and delivery estimates in MA and the other high-late states; set more realistic estimates where lateness is structural.

**Balance marketing and inventory by category (finding: category concentration)**

Spend marketing budget on health_beauty and similar high-revenue categories, and base stock decisions on units sold.

**Keep installment options visible at checkout (finding: payments)**

Because most value is paid by card and many payments use installments, make installment choices prominent.

**Investigate cancellations by month and seller (finding: payments and cancellations)**

Find out whether unavailable stock or seller behaviour drives cancellations, since each one is lost revenue.

## 10. Limitations and Assumptions

- Revenue is product price only; freight and installment interest are excluded, so it is lower than the total customers paid.
- Olist has no quantity or product-name column. Each item row is treated as one unit and products are identified by ID and category.
- Trends use January 2017 to August 2018; 2016 and September-October 2018 contain too few orders to be reliable.
- 610 products have no category and appear as 'unknown'.
- The dataset covers one marketplace over about two years, so seasonality cannot be confirmed from a single November.
- The Black Friday explanation is an inference from the daily order spike; the data has no campaign information.
- Associations (late delivery and low review scores) do not prove cause and effect.
- Repeat customers are identified by customer_unique_id; the same person with a different ID would be missed.
- Geolocation data was not needed for this analysis and is not included in the repository.